



WRITTEN PROJECT 2

BAN403

Spring, 2026

Start: 7.4.26 at 09:00

End: 17.4.26 at 12:00

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:

<https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/>

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH. Use of artificial intelligence (AI) in this exam is determined by the course coordinator. Information on AI usage and guidelines can be found on the course's Canvas page.

SUPPLEMENTARY REGULATIONS FOR HOME EXAMINATIONS

You can find supplementary regulations under the headline "Regulations"

<https://www.nhh.no/en/for-students/regulations/>

Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: 7

Number of attachments: 2

BAN403: Simulation of Business Processes

Final Project Statement

Discrete-Event Simulation of a Healthcare Service System

1. Introduction

This final project requires students to apply discrete-event simulation and business process modelling to the case *Miller Pain Treatment Center* (The case comes attached to this statement together with a excel file with the exhibits). The case provides process descriptions, activity-time observations, scheduling information and qualitative context. Students must independently interpret this information and determine how to translate it into a simulation-ready process representation.

This is a capstone assignment. Students are expected to demonstrate the ability to extract relevant information from an unstructured case, perform rigorous statistical input modelling, develop a defensible conceptual model, and evaluate system performance based strictly on simulation output.

2. Objective of the Project

The objective of the project is for students to:

- define the operational system(s) they will model based on their interpretation of the case,
- construct an appropriate conceptual representation,
- perform a complete and statistically justified input analysis,
- implement a discrete-event simulation model reflecting the conceptual design,
- design and execute simulation experiments,
- evaluate performance using statistically sound output analysis,
- formulate conclusions and recommendations supported entirely by simulation results.

The project does not prescribe model structure, resources or performance measures. Students must determine these independently, guided by the course material and textbook.

3. Required Report Structure

The report must follow the structure below. These guidelines specify required analytical components but do not prescribe specific modelling decisions.

3.1 Problem Definition and System Description (Chs. 1–5)

This section must demonstrate understanding of the existing system using the theoretical foundations from Chapters 1–5. At a minimum, students must:

- **Analyse and describe the process architecture** using frameworks from Chapters 1–2. Identify the main processes, sub-processes, flow units, and the customer perspective.
- **Explicitly discuss the design approach followed by Dr. Keith Weems:** incremental improvement or full process design. Justify using Chapter 1 concepts.
- **Use the process-oriented description from Chapter 2** to identify opportunities for improvement.
- **Identify design enablers** (Chapter 3), explaining how they support or constrain the current process.
- **Apply the design principles in Section 3.6** when analyzing weaknesses and defining desired improvements.
- **Analyze capacity and staffing issues** using principles from Chapter 5: bottlenecks, capacity–load mismatches, and implications for waiting and utilization.
- Provide a **clear and coherent description** of the existing process, system boundaries, and customer experience.

This section should be clearly divided into two subsections:

3.1a Process Architecture and Operational Analysis

(Chs. 1–3, 5)

3.1b Opportunities for Improvement

(Ch. 2, Sec. 3.6)

3.2 Conceptual Model and Simulation Design (Ch. 4)

Your conceptual model must be grounded in the **process design tools in Chapter 4**. Requirements include:

- Developing a structured process representation using appropriate tools (process maps, flow diagrams, activity networks).
- Showing how the conceptual model derives from the analysis in Section 3.1.
- Clearly stating modelling assumptions and simplifications, with justified rationale.
- Demonstrating the mapping between conceptual design and simulation implementation (entities, queues, activities, routing logic, resources).

The conceptual model must be detailed enough for the simulation to be reproducible.

Important Requirement on Simulation Software (JaamSim)

All discrete-event simulation models must be implemented in **JaamSim**. This requirement applies to:

- the AS-IS model of the system, and
- all TO-BE or alternative scenarios analysed.

Failure to submit functioning JaamSim model files will be considered a **failure to deliver the required simulation models**. Reports without matching JaamSim implementations will be assessed as incomplete.

3.3 Input Analysis (Mandatory and Rigorous)

A complete input analysis is essential for model validity. This section must demonstrate mastery of the principles discussed in Chapter 9 of the textbook. At a minimum, you must:

- **Identify all stochastic inputs** required for the model (e.g., interarrival times, service times, routing probabilities, resource schedules).
- **Extract relevant data** from the case materials and clearly describe the dataset used.

- **Fit appropriate statistical distributions** using correct methodology (histograms, empirical CDFs, parameter estimation).
- **Perform goodness-of-fit tests** (e.g., Chi-square, Kolmogorov–Smirnov) and justify distribution choices.
- **Discuss input assumptions** where the data is incomplete or ambiguous—and justify these assumptions.
- **Provide final input models** in a structured table summarizing the chosen distributions and parameters.

A weak or superficial input analysis will invalidate simulation results; this section is heavily weighted in assessment.

3.4 Output Analysis

The output analysis must follow accepted simulation methodology and include:

- experimental design and replication structure,
- chosen performance measures and their justification,
- confidence intervals for all key results,
- statistical comparison of scenarios or alternatives.

3.5 Conclusions and Recommendations (Formal Hypothesis Testing Required)

All comparisons of system performance must be supported by **formal statistical hypothesis tests**. No claims of improvement or equivalence may be made without statistical evidence.

Conclusions must be grounded entirely in simulation results and consistent with the problem definition.

3.6 Appendices, Simulation Files, and Formatting Requirements

All simulation-related materials must be submitted together with the report. This includes every file used to build, run, or validate your discrete-event simulation models. All simulation and calculation files ****must be delivered****. Failure to include these files will render numerical results invalid.

JaamSim Model Submission (Mandatory). You must submit:

- complete JaamSim project folders for the AS-IS model,
- complete JaamSim project folders for all TO-BE or alternative scenarios,
- any auxiliary files required by the model.

Models must run without errors and reproduce the results reported in the main document.

Documentation of JaamSim Implementation. You must include a short but clear documentation of each JaamSim model:

- key modelling choices and logic flow,
- mapping from conceptual model to JaamSim components,
- explanation of non-obvious assumptions.

This documentation may appear in an appendix.

Role of Appendices. Appendices may include:

- detailed input data analysis,
- additional plots or tables,
- JaamSim documentation,
- supplementary scenario results.

The ****main report must remain self-contained****: the reader should understand your reasoning and conclusions without reading the appendices.

Formatting Requirements. To ensure consistency across submissions:

- ****Font****: 12-pt serif font (Palatino or Times New Roman).
- ****Spacing****: 1.5 line spacing.
- ****Margins****: 2.5 cm on all sides.
- ****Format****: PDF only.

Page Limit. The main report must not exceed:

- ****10 pages**** (excluding appendices).

4. Information Sufficiency

The case provides sufficient information for constructing one or more discrete-event simulation models. Students are responsible for determining which elements are relevant, how to interpret available data, and which assumptions are required. All assumptions must be clearly stated and justified.